2024 AMC 10A Problem 21 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10A Problem 21. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10A Problem 21

Problem:

The numbers, in order, of each row and the numbers, in order, of each column of a 5×5 array of integers form an arithmetic progression of length 5 . The numbers in positions $(5,5),(2,4),(4,3)$, and $(3,1)$ are $0,48,16$, and 12 , respectively. What number is in position $(1,2)$?

$$\begin{bmatrix} \cdot & ? & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & 48 & \cdot \\ 12 & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & 16 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & 0 \end{bmatrix}$$

Answer Choices:

A. 19

B. 24

C. 29

D. 34

E. 39

Solution:

Let a_{ij} be the integer at row i and column j. It is given that $a_{55} = 0$, $a_{24} = 48$, $a_{43} = 16$, and $a_{31} = 12$. Suppose $a_{54} = d$. Then row 5 is $4d, 3d, 2d, d, 0$ because it is an arithmetic progression with common difference $-d$. The arithmetic progression in column 1 gives

$$a_{41} = \frac{a_{31} + a_{51}}{2} = \frac{12 + 4d}{2} = 6 + 2d$$

The arithmetic progression in column 4 gives

$$a_{44} = \frac{2a_{54} + a_{24}}{3} = \frac{2d + 48}{3} = \frac{2}{3}d + 16.$$

Row 4 gives

$$a_{43} = 16 = \frac{2a_{44} + a_{41}}{3} = \frac{\frac{4}{3}d + 32 + 6 + 2d}{3},$$

which implies $48 = \frac{10}{3}d + 38$, so $d = 3$. Filling in column 3 with common difference $16 - 6 = 10$ and column 1 with difference $12 - 12 = 0$ produces $a_{13} = 46$ and $a_{11} = 12$. Finally,

$$a_{12} = \frac{a_{13} + a_{11}}{2} = \frac{46 + 12}{2} = \text{(C)}\boxed{29}.$$

The full array looks like this:

$$\begin{bmatrix} 12 & \mathbf{29} & 46 & 63 & 80 \\ 12 & 24 & 36 & \mathbf{48} & 60 \\ \mathbf{12} & 19 & 26 & 33 & 40 \\ 12 & 14 & \mathbf{16} & 18 & 20 \\ 12 & 9 & 6 & 3 & \mathbf{0} \end{bmatrix}$$